## Trainings

## Preparatory Steps

### Aftercooled Engines

**⚠ WARNING ⚠**

**Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.**

**⚠ WARNING ⚠**

**Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.**

- If the engine is equipped with a fan, remove the fan and the fan hub. Refer to Procedure 008-036 in Section 8. (/qs3/pubsys2/xml/en/procedures/56/56-008-036-tr.html)
- Drain the cooling system. Refer to Procedure 008-018 in Section 8. (/qs3/pubsys2/xml/en/procedures/56/56-008-018-tr.html)
- Disconnect the coolant temperature sensor wire.
- Disconnect the coolant pressure sensor wire.

### LTA

**⚠ WARNING ⚠**

**Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.**

**⚠ WARNING ⚠**

**Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.**

**⚠ WARNING ⚠**

**This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.**

- If the engine is equipped with a fan, remove the fan hub. Refer to Procedure 008-036 in Section 8. (/qs3/pubsys2/xml/en/procedures/56/56-008-036-tr.html)
- Drain the cooling system. Refer to Procedure 008-018 in Section 8. (/qs3/pubsys2/xml/en/procedures/56/56-008-018-tr.html)

**Note :** On the two-stage QSK60 engine, the front intercooler **must** be removed for access to the thermostat and support housings.

- Disconnect the coolant temperature sensor wire.
- Disconnect the coolant pressure sensor wire.

QSK60 Marine Applications

**⚠ WARNING ⚠**

**Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.**

**⚠ WARNING ⚠**

**Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.**

**⚠ WARNING ⚠**

**This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.**

- If the engine is equipped with a fan, remove the fan hub. Refer to Procedure 008-036 in Section 8. (/qs3/pubsys2/xml/en/procedures/56/56-008-036-tr.html)
- Drain the cooling system. Refer to Procedure 008-018 in Section 8. (/qs3/pubsys2/xml/en/procedures/56/56-008-018-tr.html)
- Remove the sea water expansion tank. Refer to Procedure 008-052 in Section 8. (/qs3/pubsys2/xml/en/procedures/56/56-008-052-tr.html)
- Remove the sea water heat exchanger. Refer to Procedure 008-053 in Section 8. (/qs3/pubsys2/xml/en/procedures/56/56-008-053-tr.html)
- Disconnect the coolant temperature sensor wire.
- Disconnect the coolant pressure sensor wire.

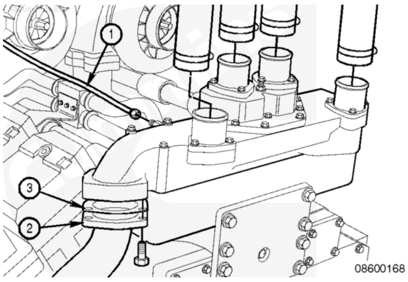
# Remove

## Aftercooled Engines

Remove the upper radiator hoses from the thermostat housing.

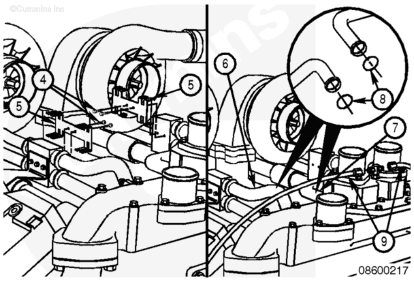
Remove the turbocharger coolant return line (1) from the rear of the thermostat housing.

Remove the bypass tube (2) and gasket (3) from thermostat housing.

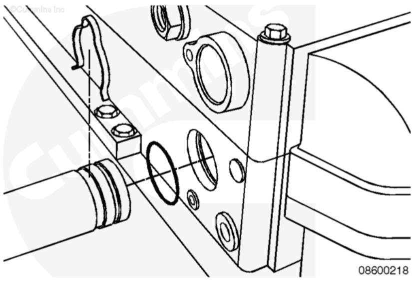



Remove the four capscrews (4) and the two retainer clips (5).

Remove the aftercooler coolant return tube (6), aftercooler coolant supply tube (7), o-rings (8), and hoses (9) from the thermostat housing support.

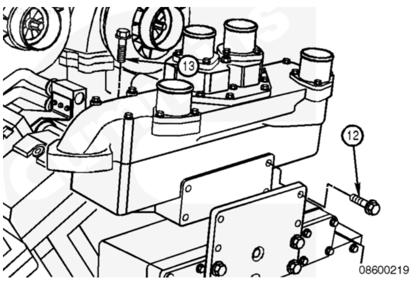



Remove the two water transfer tubes. Remove and discard the o-rings.




**⚠ WARNING ⚠**

**This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.**

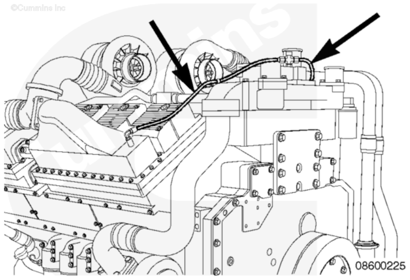



If the engine is equipped with an air compressor, disconnect the air compressor water outlet tube from the thermostat housing support.

Remove the four front mounting capscrews (12), the 10 top mounting capscrews (13), and the thermostat housing support.

## LTA

Remove the vent hoses from the thermostat housing.

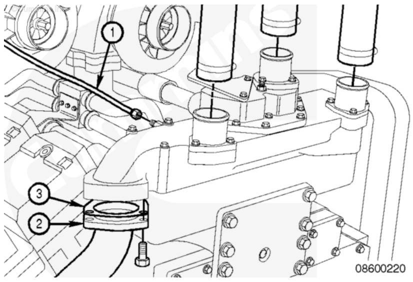



Remove the upper radiator hoses from the thermostat housing.

Remove the upper low temperature aftercooling radiator hoses from the thermostat housing.

Remove the turbocharger return lines (1) from the thermostat housing.

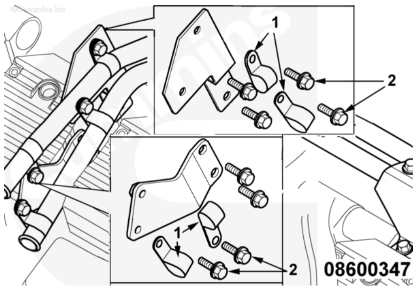
Remove the three capscrews, gasket (3), and bypass tube (2) from the thermostat housing.




### Right Bank

- Remove the capscrews (2)
- Remove the P-clips (1).

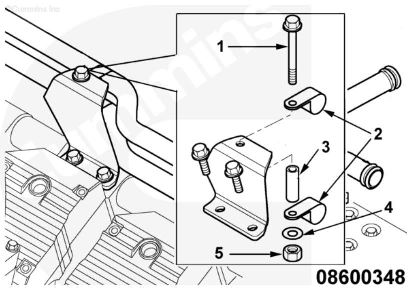
The mounting hardware attaches the upper and lower tubes to the brackets on number 5 rocker housing.




Remove the following:

- Capscrew (1)
- P-clips (2)
- Spacer (3)
- Washer (4)
- Nut (5).

The above listed mounting hardware attaches the upper and lower water tubes to the support bracket mounted on rocker housing number 2.

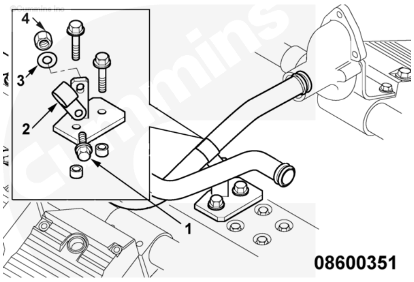



Remove the following:

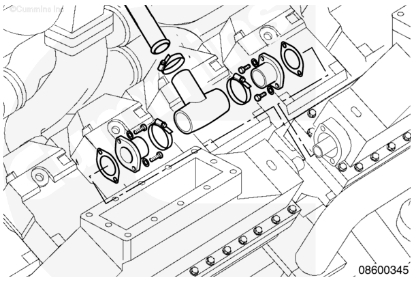
- Capscrew (1)
- P-clip (2)

- Washer (3)
- Nut (4).

The above listed mounting hardware attaches the lower water tube to the bracket mounted on the thermostat housing support.



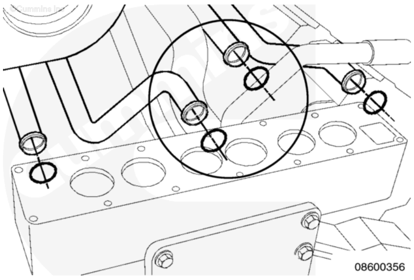

Remove the lower and upper water transfer tube from the rubber tee.




**Note : The circled area of the graphic is for location purposes only.**

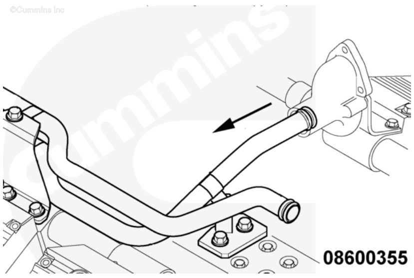
Remove the upper water transfer tube from the thermostat housing.

Remove and discard the o-rings.




Remove the lower tube from the junction block.

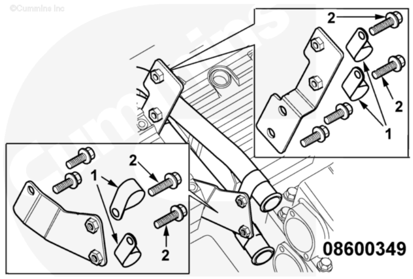
Remove and discard the o-rings.




## Left Bank

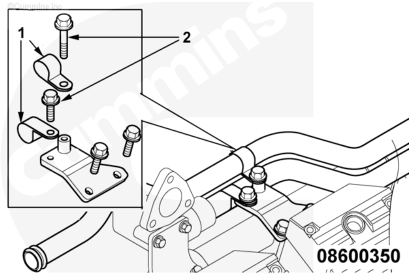
- Remove the capscrews (2)
- Remove the P-clips (1).

The mounting hardware attaches the upper and lower water transfer tubes to the brackets on rocker housing number 5.




- Remove the capscrews (2)
- Remove the P-clips (1).

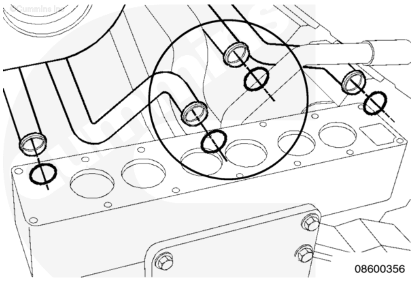
The mounting hardware attaches the upper and lower water transfer tubes to the bracket mounted on rocker housing number 2.



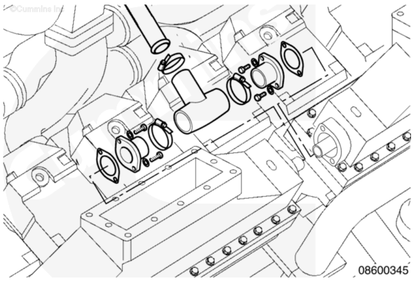

**Note : The circled area of the graphic is for location purposes only.**

Remove the upper water transfer tube from the thermostat housing.

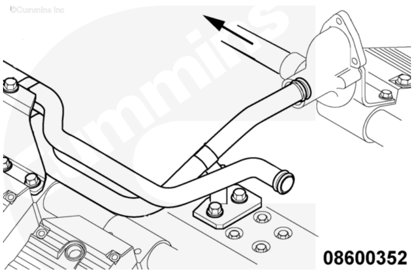
Remove and discard the o-ring.



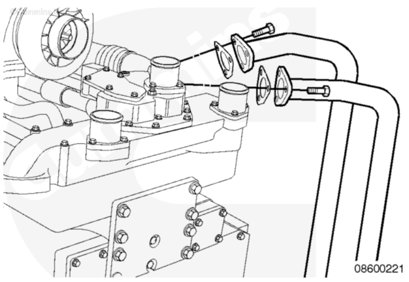

Remove the upper and lower water tubes from the rubber tee.




Remove the lower tube from the junction block.

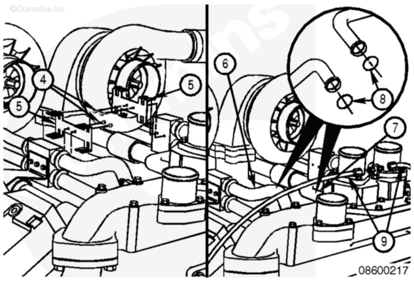



Remove the six capscrews, two gaskets, and two tubes from the thermostat housing to the low temperature aftercooling water pump.



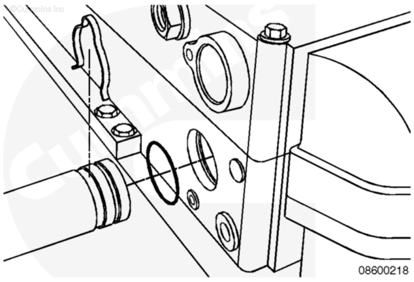

Remove the four capscrews (4) and the two retainer clips (5).

Remove the aftercooler coolant return tube (6), aftercooler coolant supply tube (7), o-rings (8), and hoses (9) from the thermostat housing support.



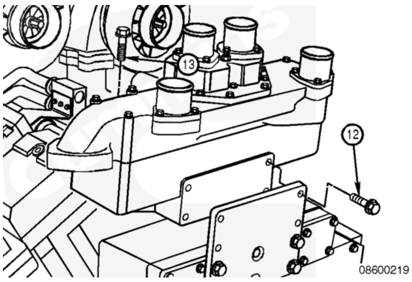

Remove the two water transfer tubes.

Remove and discard the four o-rings.




**⚠ WARNING ⚠**

**This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.**




If the engine is equipped with an air compressor, disconnect the air compressor water outlet tube from the thermostat housing support.

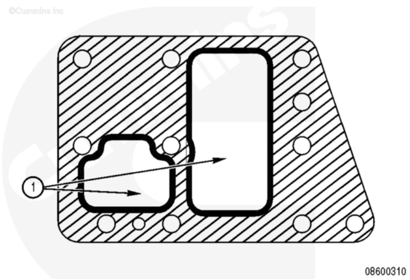
Remove the four front mounting capscrews (12), the 10 top mounting capscrews (13), and the thermostat housing support.

# Install

## Aftercooled Engines

**⚠ CAUTION ⚠**

**The thermostat housing mounting gaskets used on aftercooled engines and low temperature aftercooled engines are different. Use of the incorrect gasket on aftercooled engines will restrict coolant flow and cause high intake manifold temperatures.**



On aftercooled engines the main water pump supplies both the aftercooling and water jacket circuits.

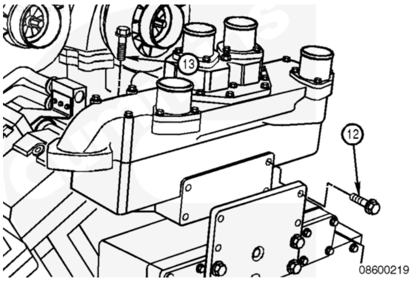
The thermostat housing gasket on aftercooled engines, illustrated in the graphic, is identified with 2 open ports (1).

**Note : Thermostat housing mounting gaskets used on low temperature aftercooled engines have only one open port.**

Install the thermostat housing mounting gasket between the cylinder block and the thermostat housing.

**⚠ WARNING ⚠**

**This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.**



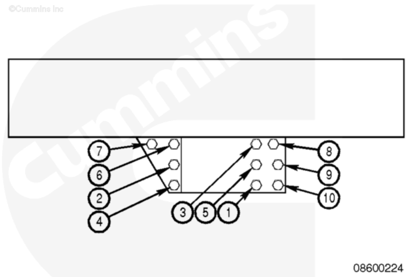

Position thermostat housing support on engine.

Install the four capscrews (12) from the front and 10 capscrews (13) from the top.

If the engine is equipped with an air compressor, connect the air compressor water outlet tube to the thermostat housing support.

Tighten the top capscrews in the pattern shown.

**Torque Value:** 80 n•m [ 59 ft-lb ]

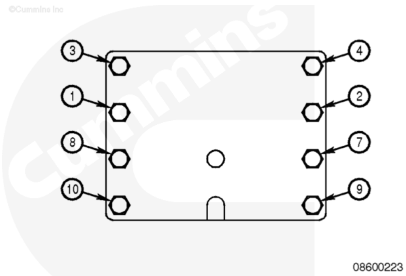



Tighten the front capscrews in the pattern shown.

**Torque Value:**

1. 195 n•m [ 144 ft-lb ]
2. 550 n•m [ 406 ft-lb ]

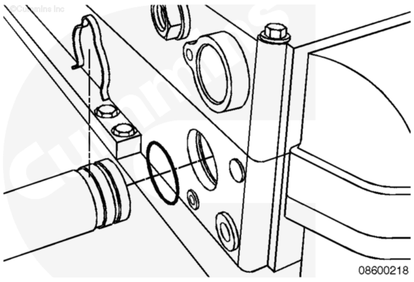
The front alignment plate is **only** used on engines without fan drives.



Use vegetable oil to lubricate the o-rings.

Install the two o-rings on the two water transfer tubes.

Install the two water transfer tubes into the thermostat housing.

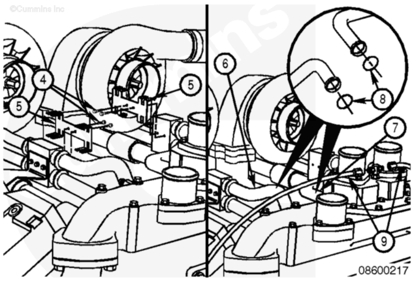



Use vegetable oil to lubricate the four o-rings (8).

Install four new o-rings (8) on the aftercooler coolant supply tubes (7) and the aftercooler coolant return tubes (6).

Install the two aftercooler coolant supply tubes (7) and the two aftercooler coolant return tubes (6) in the thermostat housing support. Install the retainer clip (5) and capscrews (4). Tighten the capscrews.

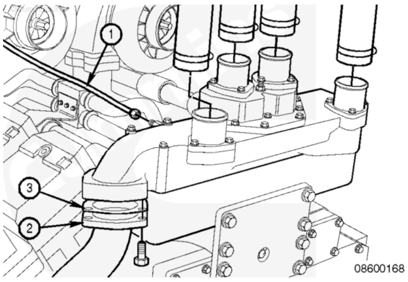
**Torque Value:** 20 n•m [ 177 in-lb ]




Install the bypass tube (2) and gasket (3) with three capscrews.

Tighten the capscrews.

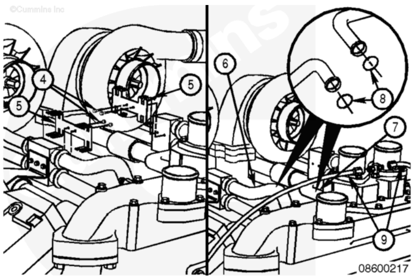
**Torque Value:** 45 n•m [ 33 ft-lb ]




Install the upper radiator hoses onto the thermostat housing.

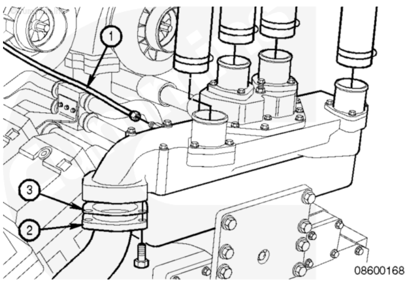
Tighten the hose clamps.

**Torque Value:** 6 n•m [ 53 in-lb ]




Install the vent lines onto the thermostat housing.

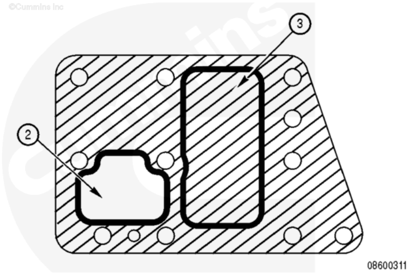
Install the turbocharger coolant return line (1) onto the thermostat housing.



LTA

**⚠ CAUTION ⚠**

**The thermostat housing mounting gaskets used on aftercooled engines and low temperature aftercooled engines are different. Use of the incorrect gasket on low temperature aftercooled engines will cause excessive pressure in the low temperature aftercooled system resulting in coolant loss from an low temperature aftercooled system surge tank overflow.**




On low temperature engines, the low temperature aftercooling circuit has a separate water pump mounted on the front cover of the engine.

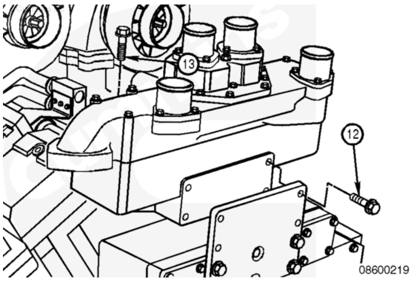
The thermostat housing mounting gasket used on low temperature aftercooled systems, illustrated in the graphic, is identified with **only** one open port (2).

**Note : The thermostat housing mounting gasket used on aftercooled engines has two open ports.**

Install the thermostat housing mounting gasket between the cylinder block and thermostat housing

**⚠ WARNING ⚠**

**This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.**



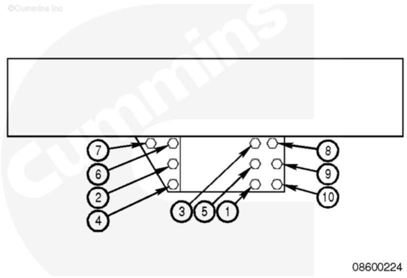

Position thermostat housing support on the engine.

Install the four capscrews (12) from the front and 10 capscrews (13) from the top.

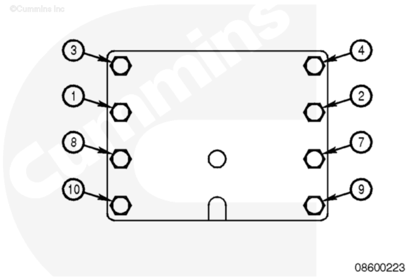
If the engine is equipped with an air compressor, connect the air compressor water outlet tube to the thermostat housing support.

Tighten the top capscrews in the pattern shown.

**Torque Value:** 80 n•m [ 59 ft-lb ]




Tighten the front capscrews in the pattern shown.

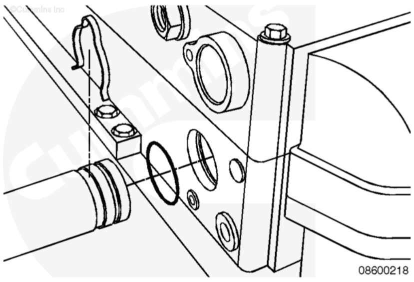



**Torque Value:**

1. 195 n•m [ 144 ft-lb ]
2. 550 n•m [ 406 ft-lb ]

**Note : Front alignment plate is only used on engines without fan drives.**

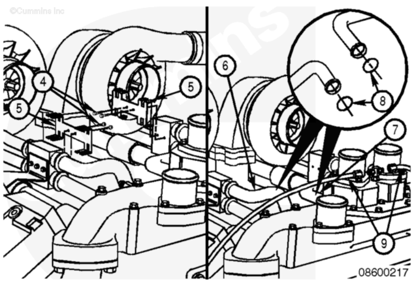
Use vegetable oil to lubricate the o-rings.




Install the o-rings on the two water transfer tubes.

Install the two water transfer tubes on the thermostat housing.

Use vegetable oil to lubricate the four o-rings (8).



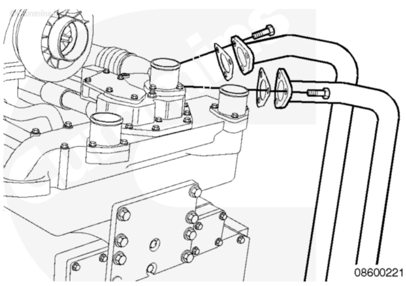

Install two new o-rings (8) on the aftercooler coolant supply tubes (7) and the aftercooler coolant return tubes (6).

Install the vent lines (9).

Install the two aftercooler coolant supply tubes (7) and the two aftercooler coolant return tubes (6) in the thermostat housing support. Install the retainer clip (5) and capscrews (4). Tighten the capscrews.

**Torque Value:** 20 n•m [ 177 in-lb ]

Install the two low temperature aftercooling coolant pump tubes, with six capscrews, into thermostat housing.

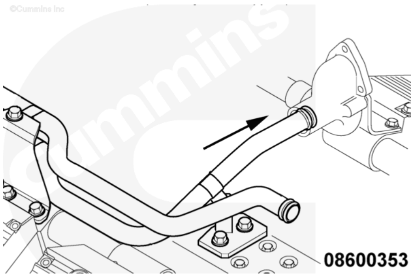



Tighten the capscrews.

**Torque Value:** 45 n•m [ 33 ft-lb ]

## ⚠ CAUTION ⚠

**Do not use petroleum based oil. Use of petroleum based oil will cause the o-ring to swell.**

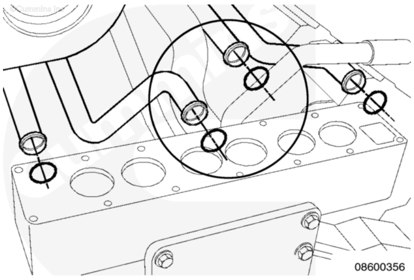



### Right Bank

- Lubricate the o-ring with vegetable oil.
- Install the o-ring onto the lower water tube.
- Fit lower water tube into junction block.

## ⚠ CAUTION ⚠

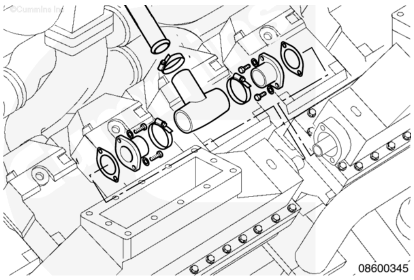
**Do not use petroleum based oil. Use of petroleum based oil will cause the o-ring to swell.**




**Note : The circled area of the graphic is for location purposes only.**

- Lubricate the o-ring with vegetable oil.
- Install the o-ring onto the tube.
- Fit the upper tube into the thermostat housing.

Fit the upper and lower water tubes into the rubber tee.

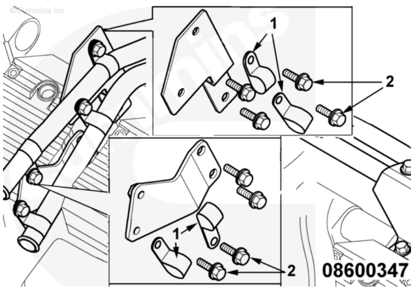



Do **not** fully tighten the p-clip capscrews until the tube is assembled onto all brackets.

Secure upper and lower tubes on the number 5 rocker housing using the following:

- P-clips (1)
- Capscrews (2).

Tighten the capscrews.




**Torque Value:** 45 n•m [ 33 ft-lb ]

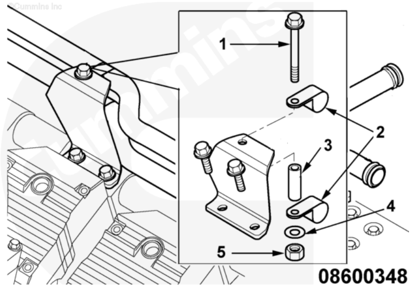
Do **not** fully tighten the p-clip capscrews until the tube is assembled onto all brackets.

Secure the upper and lower tube on the number 2 rocker housing using the following:

- Capscrews (1)
- P-clips (2)
- Spacer (3)
- Washers (4).

Tighten the capscrew.

**Torque Value:** 45 n•m [ 33 ft-lb ]

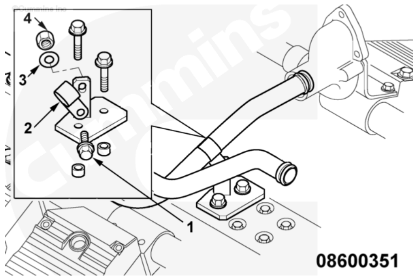



Secure the lower water pipe to the bracket on the thermostat housing using the following:

- Capscrews (1)
- P-clip (2)
- Washers (3)
- Nuts (4).

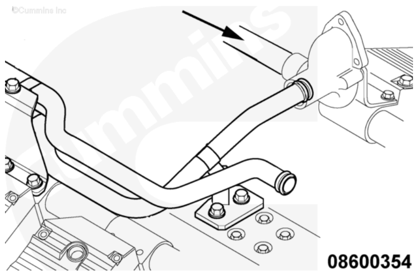
Tighten the capscrews.

**Torque Value:** 45 n•m [ 33 ft-lb ]




**⚠ CAUTION ⚠**

**Do not use petroleum based oil. Use of petroleum based oil will cause the o-ring to swell.**

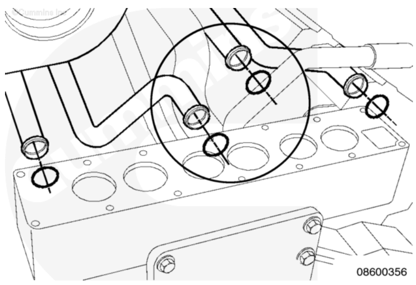



### Left Bank

- Lubricate the o-ring with vegetable oil.
- Install the o-ring onto the tube.
- Fit the lower tube into the junction block.

**⚠ CAUTION ⚠**

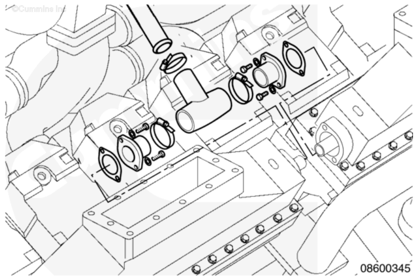
**Do not use petroleum based oil. Use of petroleum based oil will cause the o-ring to swell.**




**Note : The circled area of the graphic is for location purposes only.**

- Lubricate the o-ring with vegetable oil.
- Install the o-ring onto the tube.
- Fit the upper tube into the thermostat housing.

Fit the upper and lower water tubes into the rubber tee.

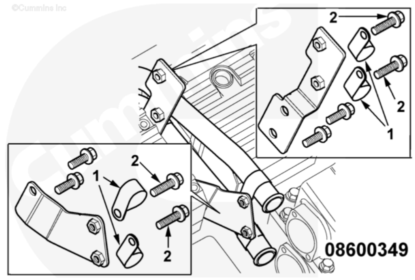



Secure both the upper and lower tubes to the brackets on the number 5 rocker housing using the following:

- P-clips (1)
- Capscrews (2).

Tighten the capscrews.

**Torque Value:** 45 n•m [ 33 ft-lb ]

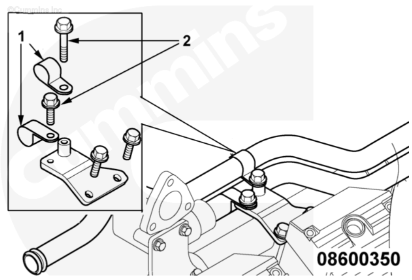



Secure the upper and lower tubes to the brackets on the number 2 rocker housing using the following:

- P-clips (1)
- Capscrews (2).

Tighten the capscrews.

**Torque Value:** 45 n•m [ 33 ft-lb ]




Install the bypass tube (2) and gasket (3) with three capscrews onto thermostat housing. Tighten the capscrews.

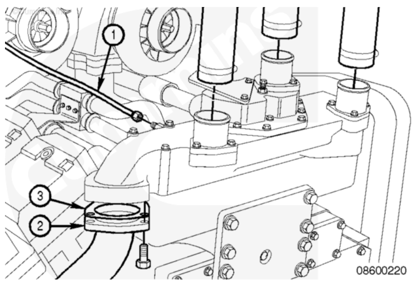
**Torque Value:** 45 n•m [ 33 ft-lb ]

Install the upper radiator hoses on the thermostat housing.

Tighten the hose clamps.

**Torque Value:** 6 n•m [ 53 in-lb ]

Install the turbocharger coolant return line (1) on the thermostat housing.




# Finishing Steps

## Aftercooled Engines

- Connect the coolant temperature sensor wire.
- Connect the coolant pressure sensor wire.

- If the fan hub was removed, install fan hub. Refer to Procedure 008-036 in Section 8. (/qs3/pubsys2/xml/en/procedures/56/56-008-036-tr.html)
- Fill the cooling system. Refer to Procedure 008-018 in Section 8. (/qs3/pubsys2/xml/en/procedures/56/56-008-018-tr.html)
- Operate the engine to 70°C [158°F] coolant temperature and check for leaks.

## LTA

- Connect the coolant temperature sensor wire.
- Connect the coolant pressure sensor wire.
- Install the front intercooler. Refer to Procedure 010-083 in Section 10. (/qs3/pubsys2/xml/en/procedures/56/56-010-083-tr.html)
- If the fan hub was removed, install the fan hub. Refer to Procedure 008-036 in Section 8. (/qs3/pubsys2/xml/en/procedures/56/56-008-036-tr.html)
- Fill the cooling system. Refer to Procedure 008-018 in Section 8. (/qs3/pubsys2/xml/en/procedures/56/56-008-018-tr.html)
- Operate the engine to 70°C [158°F] coolant temperature and check for leaks.

## QSK60 Marine Applications

- Connect the coolant temperature sensor wire.
- Connect the coolant pressure sensor wire.
- Install the sea water heat exchanger. Refer to Procedure 008-053 in Section 8. (/qs3/pubsys2/xml/en/procedures/56/56-008-053-tr.html)
- Install the sea water expansion tank. Refer to Procedure 008-052 in Section 8. (/qs3/pubsys2/xml/en/procedures/56/56-008-052-tr.html)
- If the fan hub was removed, install the fan hub. Refer to Procedure 008-036 in Section 8. (/qs3/pubsys2/xml/en/procedures/56/56-008-036-tr.html)
- Fill the cooling system. Refer to Procedure 008-018 in Section 8. (/qs3/pubsys2/xml/en/procedures/56/56-008-018-tr.html)
- Operate the engine to 70°C [158°F] coolant temperature and check for leaks.

**Last Modified: 28-Sep-2016**